

Modeling Anthropogenic CO₂ Accumulation: A Bayesian Comparison of Linear and Accelerating Trends

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Abstract

The Mauna Loa Observatory is a premier baseline site for monitoring historical and current atmospheric carbon dioxide (CO₂) concentrations. This report investigates whether the long-term increase in atmospheric CO₂ is best described by a constant linear increase or an accelerating quadratic trend. Using monthly average CO₂ data from the NOAA Global Monitoring Laboratory, we address missing values and compute annual averages to isolate the long-term trend from seasonal variations. We perform a Bayesian analysis to estimate the parameters of both a linear model and a quadratic model, with a specific focus on the acceleration parameter. The two models are then evaluated and compared using an approximate Bayes factor to determine which model better captures the underlying trend dynamics. Finally, utilizing the best-fitting model, we project the annual atmospheric CO₂ concentrations for the next decade, providing predictive estimates complete with Bayesian credible bands to quantify uncertainty.

1. Introduction

Atmospheric carbon dioxide (CO₂) is the primary anthropogenic driver of global radiative forcing, and its continuous monitoring is essential for tracking the pace of climate change. The longest continuous record of atmospheric CO₂ comes from the Mauna Loa Observatory in Hawaii, maintained by the NOAA Global Monitoring Laboratory (GML), and commonly known as the Keeling Curve. This record is the benchmark dataset for studying the long-term evolution of atmospheric CO₂.

A central question in climate science is not only whether CO₂ concentrations are increasing, but how the rate of increase is itself evolving. A constant linear growth rate would imply that annual anthropogenic emissions have remained roughly stable over time, whereas an accelerating, quadratic growth rate would indicate that emissions are still rising, compounding the rate of atmospheric accumulation. Distinguishing between these two regimes has direct implications for emission-reduction targets and for the reliability of concentration forecasts used in climate policy.

In this report, we use the monthly CO₂ record from Mauna Loa to formally compare a linear trend model against a quadratic, accelerating trend model within a Bayesian framework. After describing

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and preprocessing the dataset, we estimate the parameters of both models, evaluate the evidence for acceleration using an approximate Bayes factor, and use the best-supported model to produce decadal projections of atmospheric CO₂ with full posterior uncertainty.

1.1. Scientific Context

Atmospheric CO₂ is regulated by the balance between anthropogenic emissions (fossil fuel combustion, land-use change) and natural sinks (ocean uptake, terrestrial photosynthesis). Since continuous monitoring began in 1958, the Mauna Loa record has shown a persistent rise from approximately 315 ppm to over 420 ppm, superimposed on a strong seasonal cycle driven by the Northern Hemisphere growing season.

While the existence of an upward trend is undisputed, its functional form is of practical interest: a purely linear fit corresponds to a constant annual increment, whereas curvature in the trend would indicate that the rate of increase is itself growing, consistent with the continued rise in global fossil fuel emissions reported in recent decades. Capturing this curvature correctly matters for extrapolation, since a linear model will systematically underestimate future concentrations if the underlying process is in fact accelerating.

A Bayesian approach is well suited to this comparison: it provides a principled framework both for estimating the acceleration parameter together with its uncertainty, and for directly weighing the evidence for the linear and quadratic hypotheses through an approximate Bayes factor, rather than relying solely on goodness-of-fit criteria. The resulting posterior predictive distribution also yields credible bands for future CO₂ projections, which communicate forecast uncertainty more naturally than a single point estimate.

1.2. Objectives

2. Data Description and Preprocessing

The dataset used in this analysis consists of monthly average atmospheric CO₂ concentrations measured at the Mauna Loa Observatory, Hawaii, from 1958 to 2026. The data is sourced from the NOAA Global Monitoring Laboratory and is available in a TXT format. Each record includes the date of measurement and the corresponding CO₂ concentration in parts per million (ppm).

The dataset is structured as follows:

- **year:** The year of the measurement.
- **month:** The month of the measurement.
- **decimal_date:** The decimal representation of the date (e.g., 2020.5 for June 2020).
- **monthly_average:** The average CO₂ concentration for that month in ppm.
- **deseasonalized:** The monthly average CO₂ concentration with the seasonal component removed.
- **num_days:** The number of days in the month used for averaging.
- **st_dev:** The standard deviation of the daily CO₂ measurements for that month.
- **unc_mon_mean:** The uncertainty in the monthly mean CO₂ concentration.

2.1. Dataset Overview

Table 1: Summary statistics of the Mauna Loa CO2 dataset

Variable	Missing	Mean	SD	Min	Median	Max
year	0	1991.75	19.69	1958.00	1992.00	2026.00
monthly_average	0	361.02	33.19	312.42	356.26	431.12
deseasonalized	0	361.02	33.13	314.44	356.42	428.72
num_days	195	25.48	4.11	2.00	26.00	31.00
st_dev	196	0.51	0.20	0.15	0.48	1.31
unc_mon_mean	194	0.19	0.08	0.00	0.18	0.58

As shown in Table 1, the dataset contains some missing values, we will clean the dataset so that we can perform our analysis on a complete dataset.

2.2. Data Cleaning

The dataset contains some missing values, which are we dropped all the rows with missing values limiting our analysis to the period from 1974 to 2026.

Table 2: Summary statistics of the Mauna Loa CO2 dataset after removing missing values

Variable	Missing	Mean	SD	Min	Median	Max
year	0	1999.90	15.01	1974.00	2000.00	2026.00
monthly_average	0	373.36	28.38	327.28	369.80	431.12
deseasonalized	0	373.36	28.30	329.79	369.40	428.72
num_days	0	25.52	4.00	7.00	26.00	31.00
st_dev	0	0.51	0.20	0.15	0.48	1.31
unc_mon_mean	0	0.20	0.08	0.06	0.18	0.58

After cleaning the dataset, we can see that the statistics of the dataset have changed, with the mean and standard deviation of the variables being slightly different from the original dataset. Also, the main variables of interest, **monthly_average** and **deseasonalized**, were not affected by the missing values, so by removing the missing values we lose valuable information.

That's why we decided to replace the missing values with the mean of the variable, monthly average, so the inferred values are more realistic and the analysis is not affected by the missing values.

Table 3: Summary statistics of the Mauna Loa CO2 dataset after replacing missing values

Variable	Missing	Mean	SD	Min	Median	Max
year	0	1991.75	19.69	1958.00	1992.00	2026.00
monthly_average	0	361.02	33.19	312.42	356.26	431.12
deseasonalized	0	361.02	33.13	314.44	356.42	428.72
num_days	0	25.33	3.64	2.00	26.00	31.00
st_dev	0	0.51	0.18	0.15	0.50	1.31

Table 3: Summary statistics of the Mauna Loa CO2 dataset after replacing missing values

Variable	Missing	Mean	SD	Min	Median	Max
unc_mon_mean	0	0.19	0.07	0.00	0.19	0.58

2.3. Aggregation

While the original dataset is what the analysis will focus on, there might be useful to compute other quantities in order to understand better the data, those quantities are:

- **yearly_average**: The average CO2 concentration for each year, computed by averaging the monthly averages for that year (See Figure 3).
- **daily_average**: The average CO2 concentration for each number of days measured in a month, computed by averaging the monthly averages for each unique value of `num_days` (See Figure 4).
- **seasonal_deviation**: The deviation of the monthly average CO2 concentration from the deseasonalized value, computed as `monthly_average - deseasonalized` (See Figure 5).

2.4. Exploratory Visualization

In this section, we will explore the data through various visualizations to understand the underlying patterns and relationships between the variables.

2.4.1. Trend Analysis

From Figure 2, we observe a clear upward trend in atmospheric CO2 levels over time, with the data showing a consistent increase in annual average CO2 concentrations. This visual evidence suggests that the long-term trend may not be purely linear, prompting us to consider both linear and accelerating models for further analysis.

In Figure 3, we can see the yearly average CO2 concentrations, which also show a clear upward trend over time, reinforcing the observation from the monthly data. We can also examine whether there is a relationship between the number of days measured in a month and the monthly average CO2 concentration,

which would mean that the more days measured in a month, the higher the monthly average CO2 concentration, this could be due to the fact that months with more measurements are more likely to capture higher CO2 levels, especially during periods of rapid increase. In Figure 4, we observe that the monthly average CO2 concentration is uniformly distributed across different values of `num_days`, suggesting that the number of days measured in a month does not have a significant impact on the monthly average CO2 concentration.

2.4.2. Seasonal Deviation Analysis

In this analysis the data can be understood as a time series with a seasonal component, some months of the year have higher CO2 concentrations than others, this is due to the fact that during the summer months, plants are actively photosynthesizing and absorbing CO2 from the atmosphere, and people are more likely to be outside and emitting CO2, while during the winter months, plants are dormant and less CO2 is absorbed, and people are more likely to be inside and emitting less CO2.

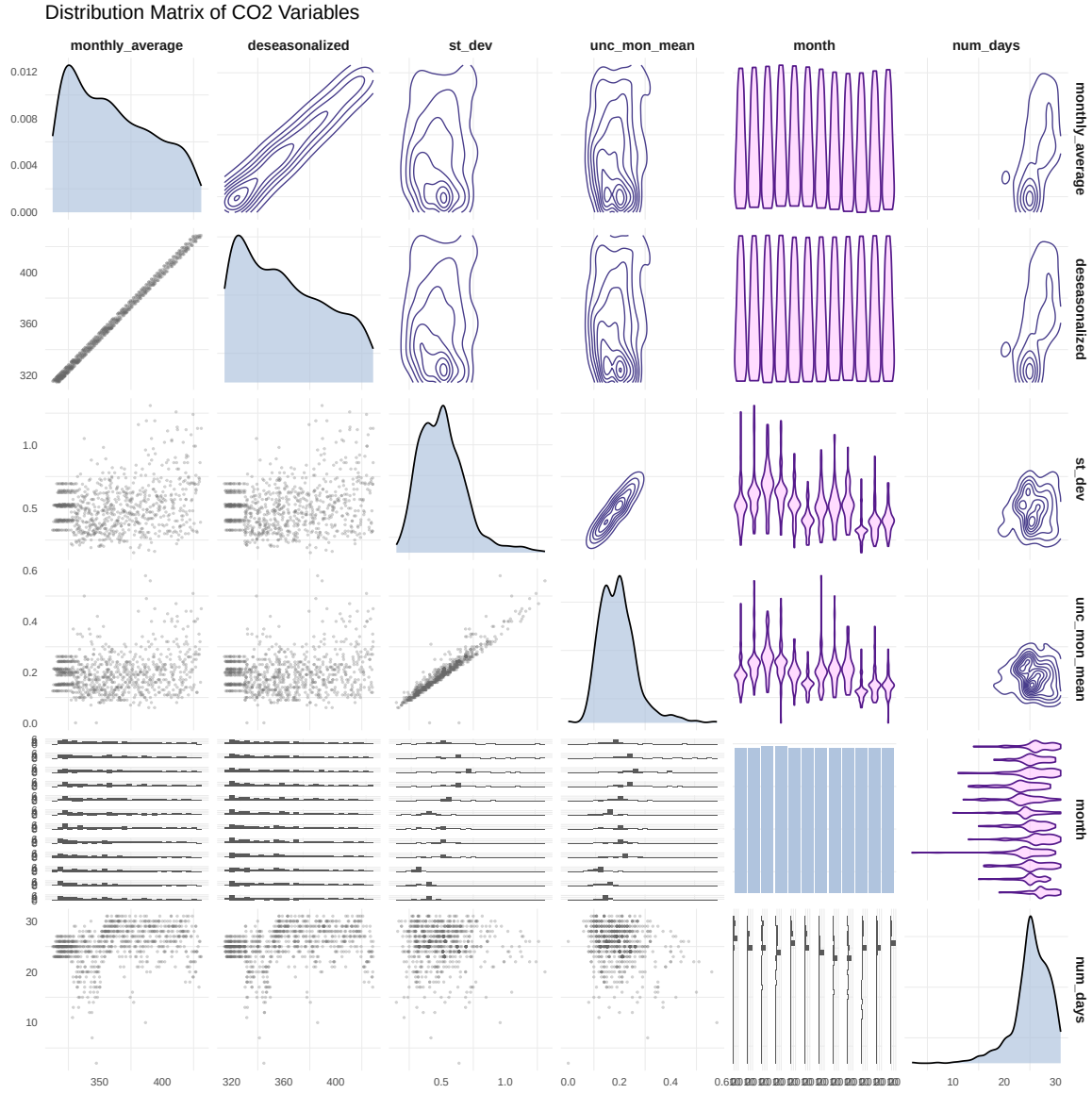


Figure 1: Distribution and correlation of the variables in the dataset. The diagonal panels show the distribution of each variable, while the upper panels show the correlation coefficients between pairs of variables. The lower panels display scatter plots.

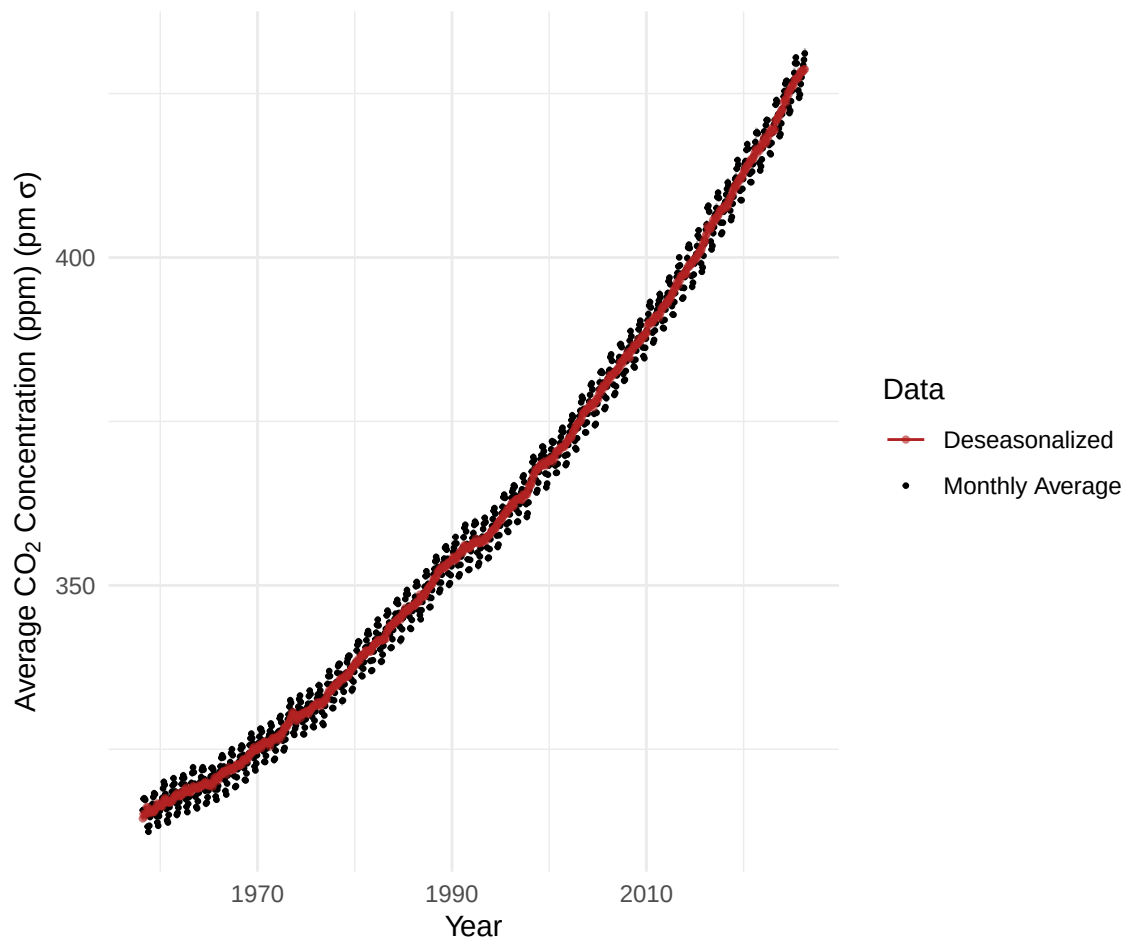


Figure 2: Monthly average CO₂ concentrations. The plot shows a clear upward trend in atmospheric CO₂ levels over time.

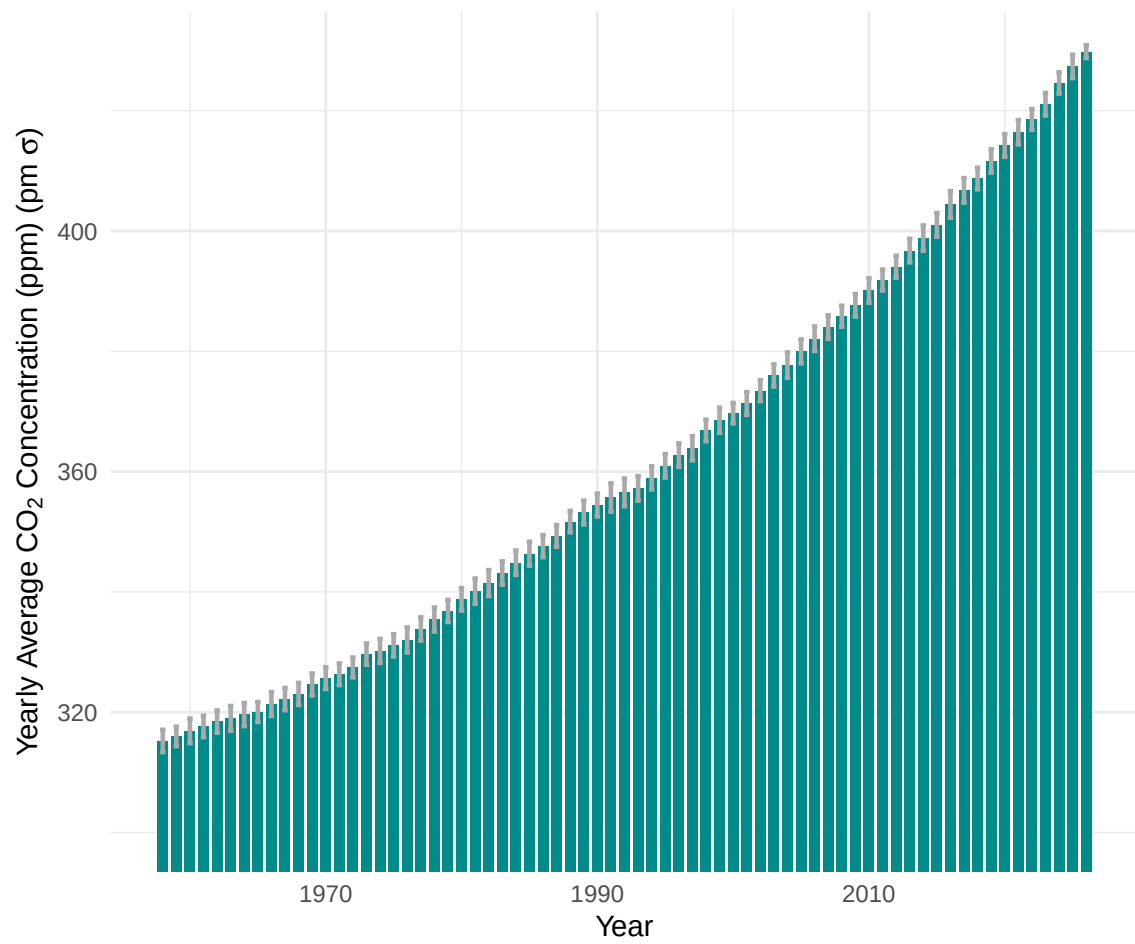


Figure 3: Yearly average CO₂ concentrations. The plot shows a clear upward trend in atmospheric CO₂ levels over time.

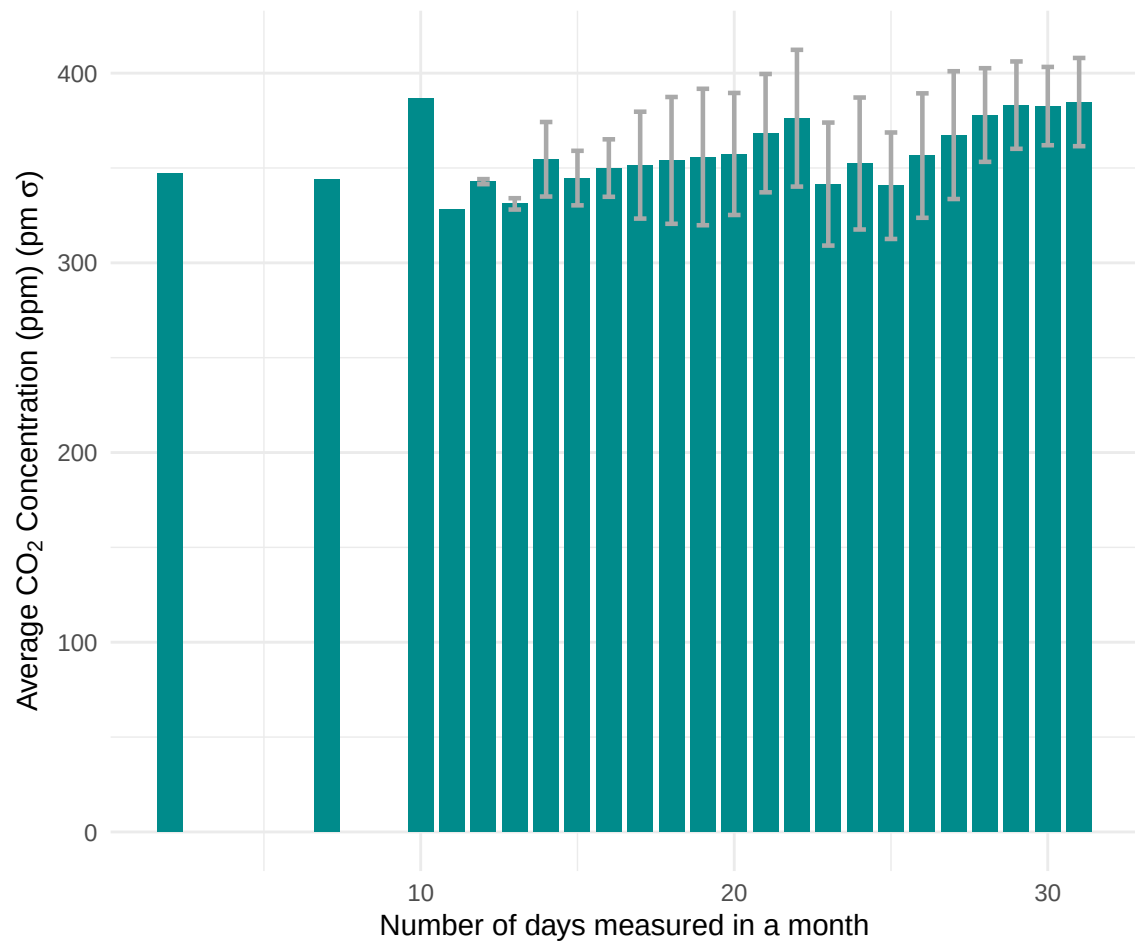


Figure 4: Daily average CO₂ concentrations. The plot shows a clear upward trend in atmospheric CO₂ levels over time.

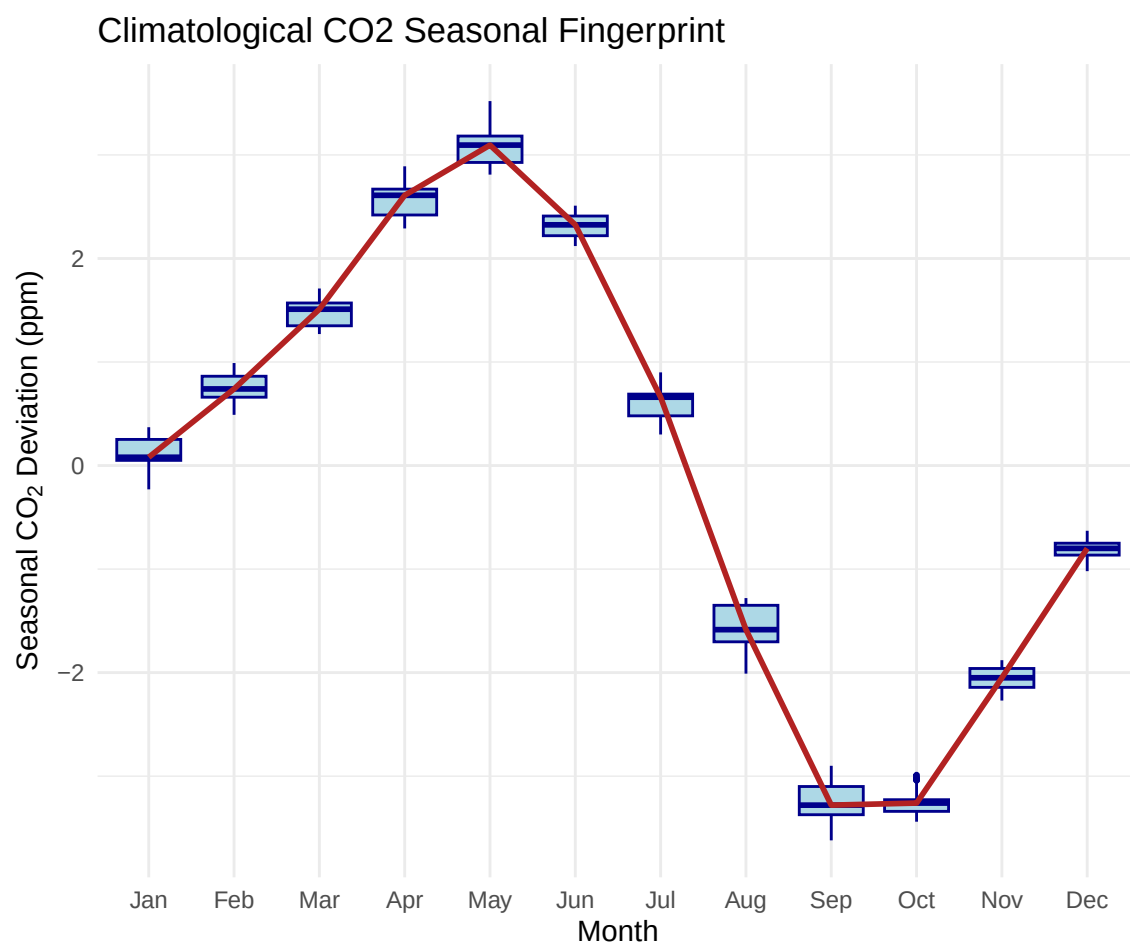


Figure 5: Box plot of seasonal deviations in CO₂ concentrations.

In figure Figure 5, we can see the distribution of seasonal deviations in CO2 concentrations across different months. The box plot illustrates the variability in CO2 levels, with certain months exhibiting higher median values and greater spread, indicating that seasonal factors significantly influence atmospheric CO2 concentrations.

2.4.3. Uncertainty Analysis

As the goal of this analysis is to make a bayesian inference model, it is important to understand the uncertainty in the data, this can be done by looking at the standard deviation of the daily measurements and the uncertainty in the monthly mean.

Data Quality and Heteroskedasticity Assessment

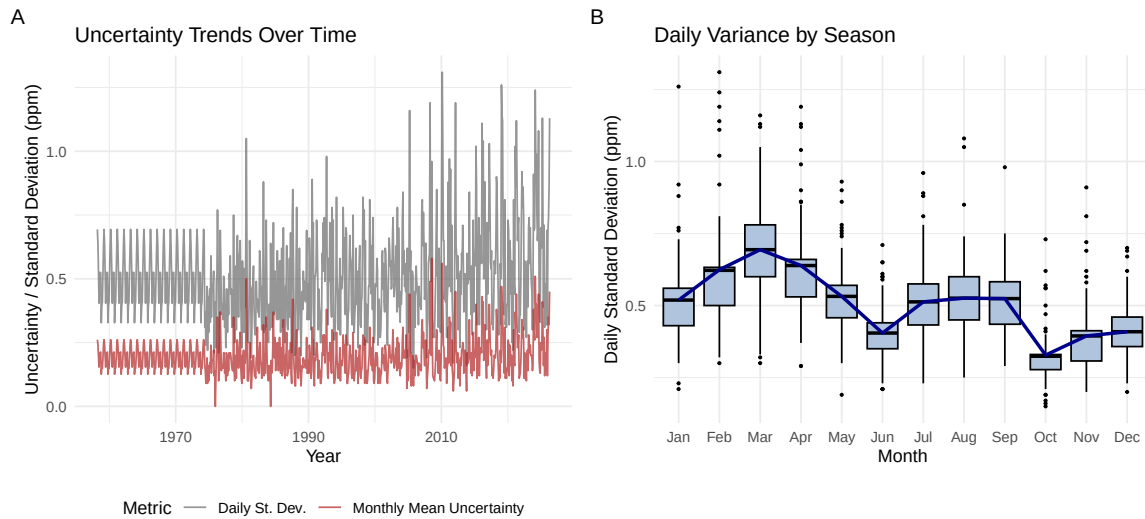


Figure 6: Uncertainty analysis of the CO2 measurements. The left panel shows the trends of daily standard deviation and monthly mean uncertainty over time, while the right panel shows the distribution of daily standard deviation across different months.

In Figure 6, we can see that the monthly mean uncertainty is generally lower and more stable than the daily standard deviation, reducing the impact of short-term fluctuations and providing a more reliable measure of the long-term trend in atmospheric CO2 concentrations. The right panel shows that the daily standard deviation varies across different months, with some months exhibiting higher variability than others, indicating that seasonal factors also influence the uncertainty in CO2 measurements.

2.4.4. Correlation Analysis

As expected from prior plots, most of the variables in the dataset are highly correlated, as there is a clear upward trend in CO2 concentrations over time.

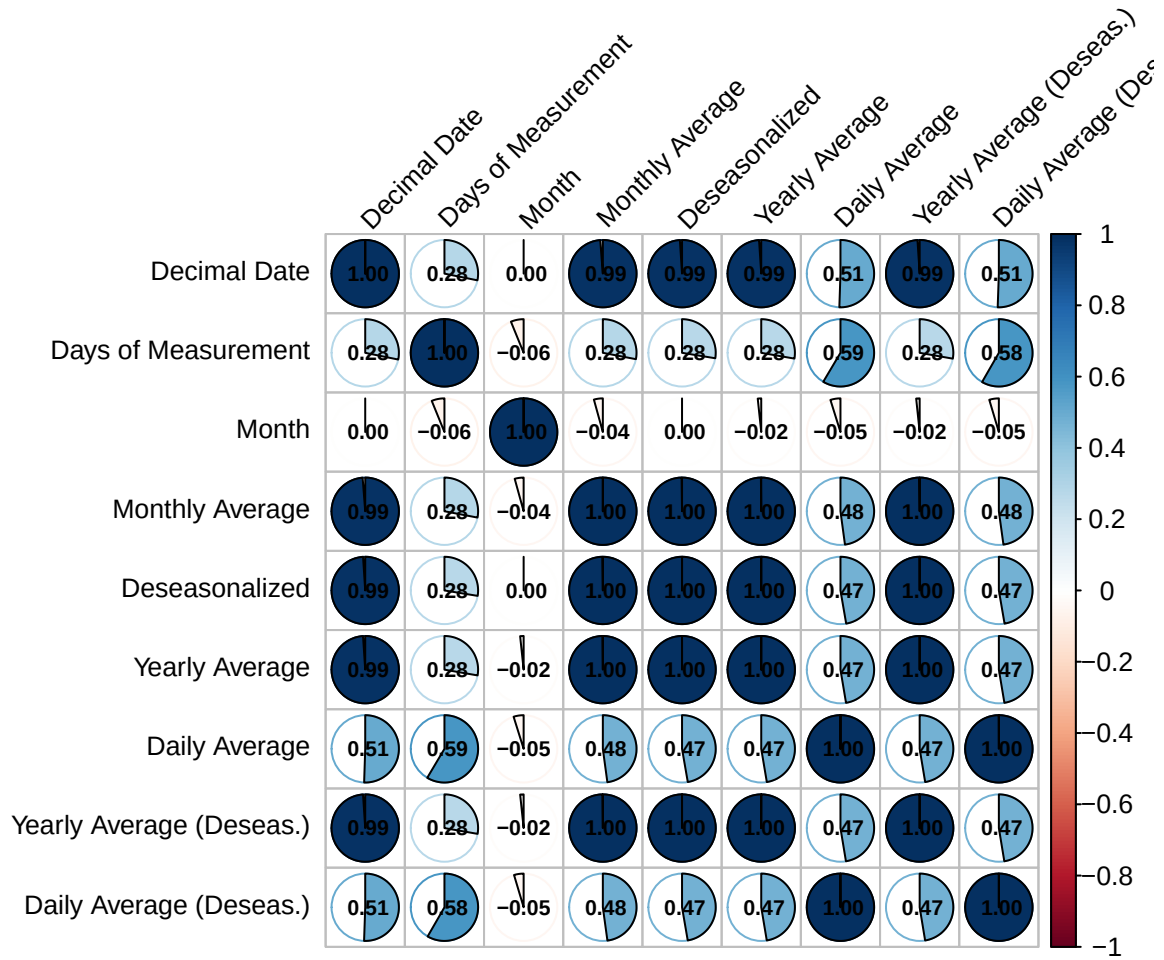


Figure 7: Correlation plot of the variables in the dataset

The daily average is lowly correlated as we show in Figure 4, as is mostly uniformly distributed across different values of `num_days`, while the other variables are highly correlated, as they are all derived from the monthly average CO2 concentration.

3. Bayesian Methodology and Models

We are considering two different models that could explain our dataset. The first model considers the data to be sampled from a linear function with the addition of noise:

$$y_t = \alpha + \beta t + \epsilon_t$$

The second model considers the data to be sampled in a similar way, but with the addition of a quadratic term (that we can consider as an acceleration):

$$y_t = \alpha + \beta t + \gamma t^2 + \epsilon_t$$

To infer an estimation of our parameters we have tried different methods: MAP estimator and STAN method. Each one of the methods is presented in the following paragraphs. In this analysis we have used the data averaged over one year. Moreover, we have rescaled time:

$$t_{scaled} = \frac{t - \bar{t}}{s_t}$$

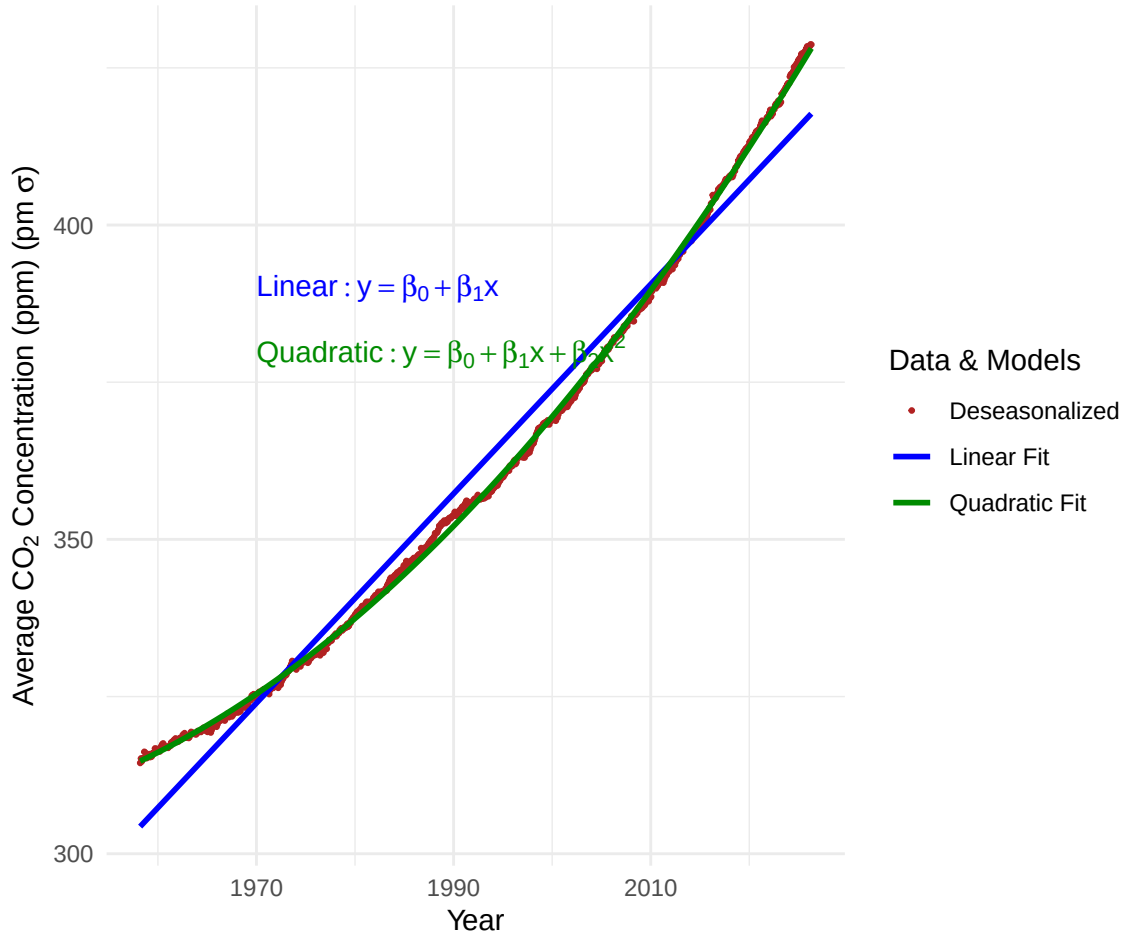


Figure 8: Deseasonalized CO2 concentrations and Bayesian models fit. The plot shows the deseasonalized CO2 concentrations along with the fitted lines from both the linear and quadratic models, illustrating the differences in their respective fits to the data.

In Figure 8, we can see the deseasonalized CO2 concentrations along with the fitted lines from both the linear and quadratic models. A priori, the quadratic model seems to capture the upward curvature in the data better than the linear model.

3.1. Model 1: Linear Trend

The linear trend model assumes that the annual CO2 concentrations increase at a constant rate over time, described by the equation $y_t = \alpha + \beta t + \epsilon_t$. This model is the simplest representation, where α represents the baseline concentration and β represents the constant annual increase rate.

3.2. Model 2: Quadratic Trend

The quadratic trend model includes an additional acceleration term, described by $y_t = \alpha + \beta t + \gamma t^2 + \epsilon_t$. The parameter γ captures the acceleration, allowing the rate of increase to grow over time. This model is more flexible and can capture non-linear patterns in the CO2 accumulation.

3.3. Inference Strategy

For parameter estimation, we employ two complementary approaches. First, we use Maximum A Posteriori (MAP) estimation to obtain point estimates of the parameters. Second, we use Hamiltonian Monte Carlo via the STAN framework to obtain full posterior distributions, enabling us to quantify uncertainty and perform rigorous model comparison.

4. Results and Inference

4.1. Parameter Estimation

4.1.1. Inference of parameters via MAP

For the MAP estimate we have maximized the following joint posteriors:

$$\log(p_{lin}(\theta|y)) = \log(p(\alpha)) + \log(p(\beta)) + \log(p(\sigma)) + \log(p_{lin}(y|\alpha, \beta, \sigma))$$

$$\log(p_{quad}(\theta|y)) = \log(p(\alpha)) + \log(p(\beta)) + \log(p(\gamma)) + \log(p(\sigma)) + \log(p_{quad}(y|\alpha, \beta, \gamma, \sigma))$$

We are considering the following priors:

$$\begin{aligned}\alpha &\propto N(350, 50^2) \\ \beta &\propto N(0, 10^2) \\ \gamma &\propto N(0, 5^2) \\ \sigma &\propto \text{Exp}(1)\end{aligned}$$

and the likelihoods are in the following form:

$$\log(p_{lin}(y|\alpha, \beta, \sigma)) = \sum_t \log(N(y_t|\alpha + \beta t, \sigma^2))$$

$$\log(p_{quad}(y|\alpha, \beta, \gamma, \sigma)) = \sum_t \log(N(y_t|\alpha + \beta t + \gamma t^2, \sigma^2))$$

In this way we find the point estimation given by:

$$\hat{\theta}_{lin} = \arg \max_{\theta} [\log(p_{lin}(y|\alpha, \beta, \sigma))]$$

$$\hat{\theta}_{quad} = \arg \max_{\theta} [\log(p_{quad}(y|\alpha, \beta, \gamma, \sigma))]$$

where $\hat{\theta}_{lin} = (\hat{\alpha}_{lin}, \hat{\beta}_{lin}, \hat{\sigma}_{lin})$ and $\hat{\theta}_{quad} = (\hat{\alpha}_{quad}, \hat{\beta}_{quad}, \hat{\gamma}_{quad}, \hat{\sigma}_{quad})$.

The maximization process has been done with the method “BFGS” and gave the following results:

Table 4: MAP estimates for the linear model

parameter	map	converged
alpha	361.58	TRUE
beta	33.46	TRUE
sigma	4.71	TRUE

Table 5: MAP estimates for the quadratic model

parameter	map	converged
alpha	356.20	TRUE
beta	33.56	TRUE
gamma	5.46	TRUE
sigma	0.74	TRUE

From Table 4 and Table 5 we notice a substantial drop in the parameter σ between the two models. This suggests that the quadratic model has the capacity to absorb a large part of the data’s deviations, whereas the linear model leaves more residual variance unexplained.

In the context of Bayesian inference, MAP estimators provide only a point estimation dependent on the local behavior of the posterior near its maximum, regardless of spread, skewness, or multimodality. For this reason, we should consider methods that capture the full joint posterior distribution. With this purpose we present the results from STAN.

4.1.2. Inference of parameters via STAN

For both models we have run 4 chains using STAN. The results are as follows:

Table 6: STAN inference results for the quadratic model

variable	mean	median	sd	mad	q5	q95	rhat	ess_bulk	ess_tail
alpha	356.198	356.198	0.140	0.137	355.966	356.429	1.001	4983.650	5148.797
beta	33.563	33.563	0.094	0.092	33.411	33.720	1.001	6813.958	5013.988

Table 6: STAN inference results for the quadratic model

variable	mean	median	sd	mad	q5	q95	rhat	ess_bulk	ess_tail
gamma	5.458	5.457	0.106	0.105	5.288	5.633	1.001	5000.091	5141.394
sigma	0.771	0.767	0.067	0.067	0.670	0.887	1.000	7052.314	6009.784

For both models, autocorrelation should be low, as can be noticed from the large ESS values, and all 4 chains are converging to the same stationary distribution, indicated by $\hat{R} = 1.00$.

Using posterior predictive check (PPC) plots, we can compare the kernel density estimate (KDE) of the real data with those generated by STAN. The quadratic model clearly reproduces the real distribution more accurately than the linear model:

Let's visualize the two models' predictions against the observed data:

4.2. Model Comparison

In the parameter inference section, we already observed that the quadratic model is a better approximation of the data. Now we formalize this comparison using two methods: the Bayes factor and the Leave-One-Out (LOO) cross-validation method.

4.2.1. Model comparison with Bayes Factor

The Bayes factor is the ratio between the marginal likelihoods of the two models:

$$BF = \frac{p(y | M_1)}{p(y | M_0)}$$

where $p(y | M_k) = \int p(y | \theta, M_k) p(\theta | M_k) d\theta$.

Since this integral is typically intractable, we use Laplace's approximation, which expands the log-posterior around the MAP estimate $\bar{\theta}$:

$$\log p(y | M_k) \propto \log p(\bar{\theta} | y) + \frac{d}{2} \log(2\pi) - \frac{\log \det(H)}{2}$$

where H is the negative of the log-posterior's Hessian.

Table 7: Bayes factor comparing quadratic vs. linear models

Log marginal likelihood (linear):	-224.9434
Log marginal likelihood (quadratic):	-101.3877
Log10 BF (quad vs lin):	53.65954

Since $\log_{10}(BF) = 53.66 \gg 1$, we can definitively conclude that the quadratic model is far more appropriate to represent our data.

MAP estimates: common parameters across models

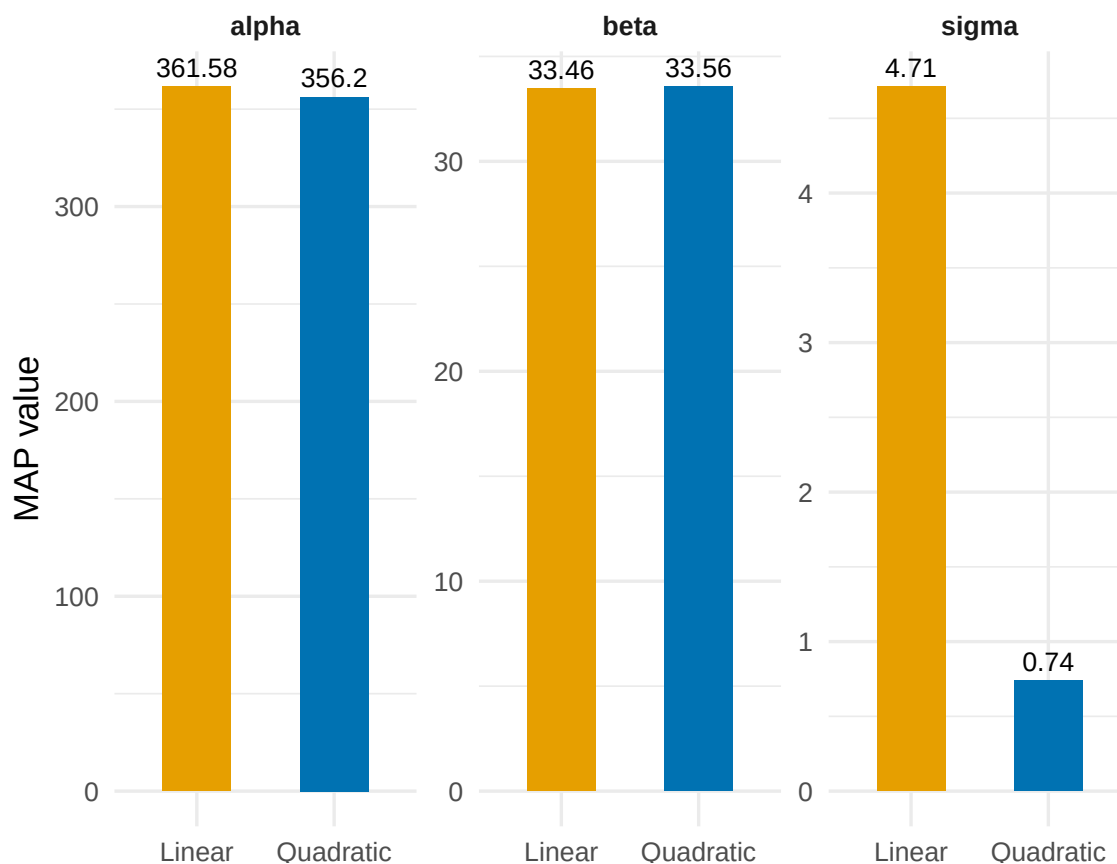


Figure 9: Comparison of MAP parameters between linear and quadratic models. This plot illustrates how the addition of a quadratic term impacts parameter inference, particularly the dramatic reduction in residual standard deviation.

MAP estimates: linear vs quadratic CO... trend

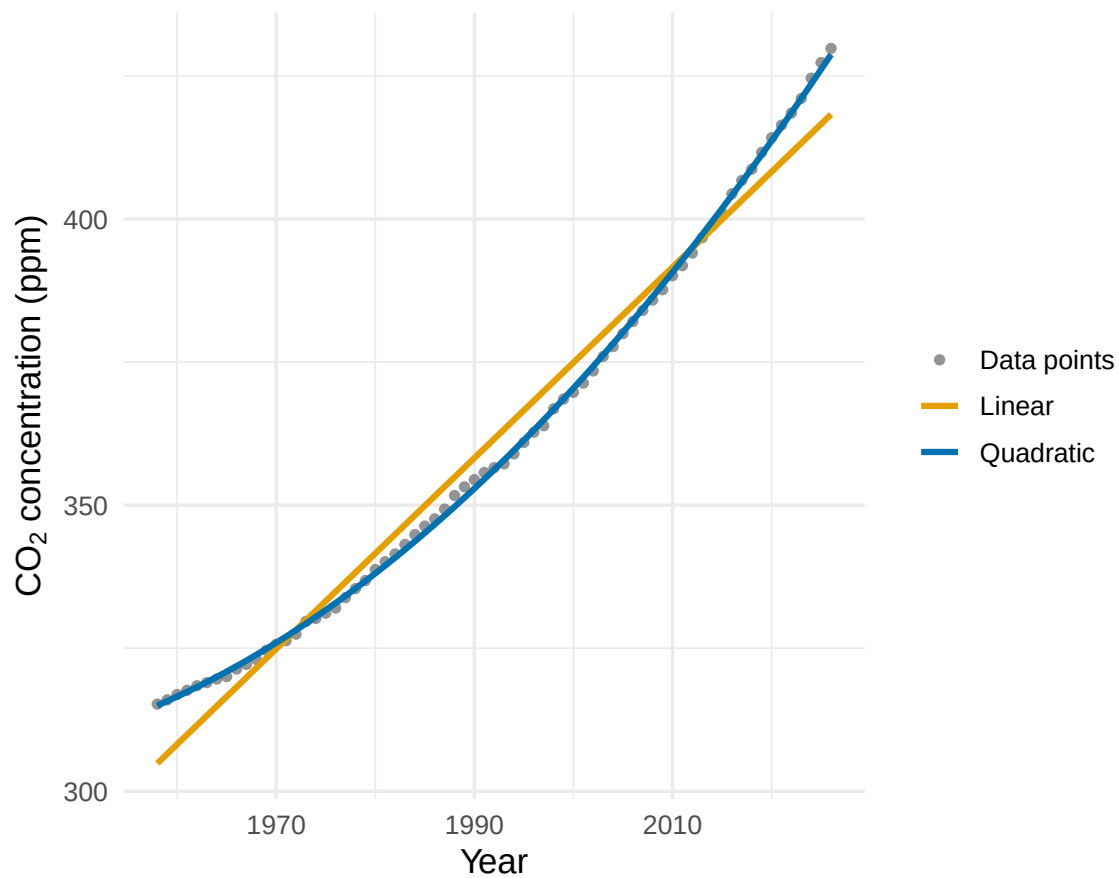


Figure 10: MAP model fits versus observed data. The linear model (orange) and quadratic model (blue) are compared against the yearly average CO₂ concentrations. The quadratic fit clearly follows the data more efficiently.

Posterior predictive check: linear vs quadratic CO₂ trend

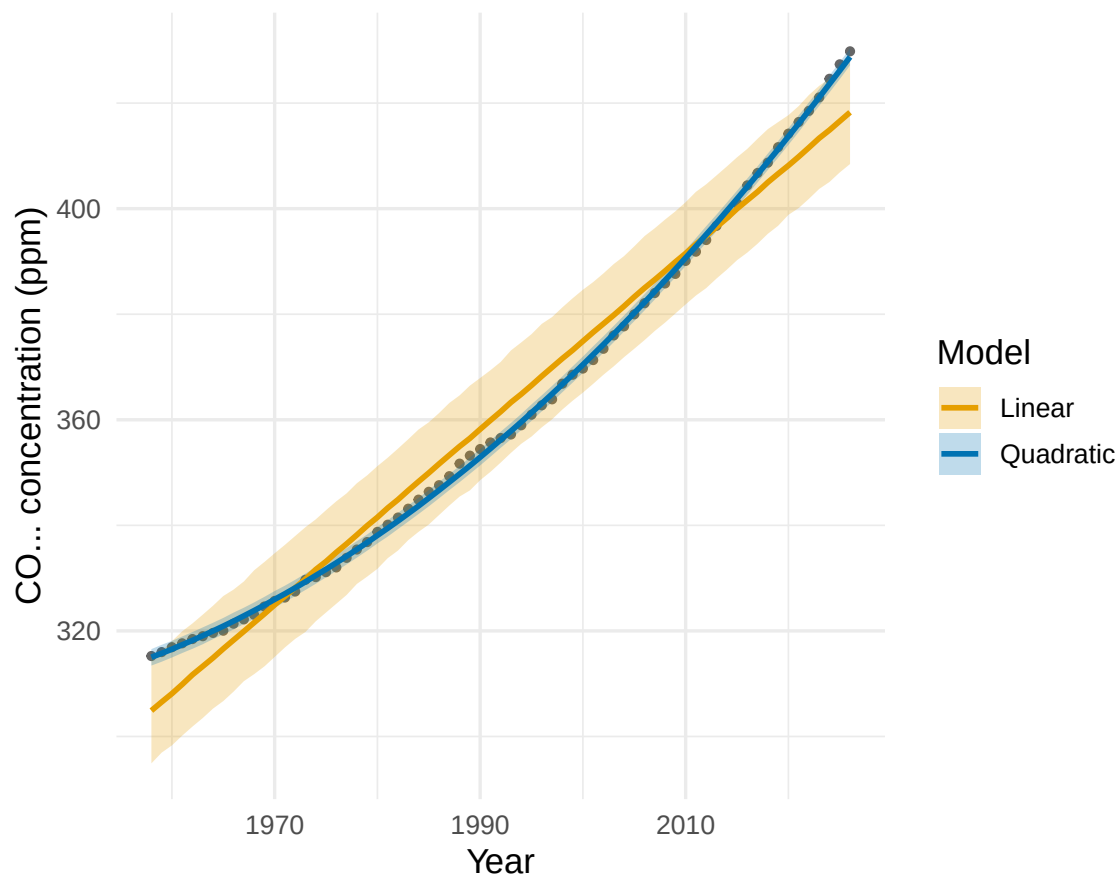


Figure 11: Yearly average CO₂ concentrations versus model predictions. The linear fit (orange) and quadratic fit (blue) are shown with 95% uncertainty bands. The quadratic model clearly captures the data better.

4.2.2. Model comparison with LOO method

The Leave-One-Out (LOO) cross-validation method assesses how well our model predicts unseen data by calculating the expected log predictive density (ELPD):

$$\text{elpd} = \sum_{i=1}^n \int \log p(y_i^* | y_{-i}) p_t(y_i^*) dy_i^*$$

where $p(y_i^* | y_{-i})$ are the log posterior predictive densities for y_i^* when the model was fitted without observation i .

The models are compared via:

$$\Delta \hat{ELPD} = \hat{ELPD}_{lin} - \hat{ELPD}_{quad}$$

Table 8: LOO model comparison table

	Model	ELPD_LOO	SE_ELPD	LOOIC	SE_LOOIC
model2	Linear	-81.60	5.60	163.19	11.21
model1	Quadratic	-210.46	5.62	420.93	11.25

The ratio $|\Delta \hat{ELPD}|/SE \approx 16.3$ decisively confirms that the quadratic model has much greater predictive ability compared to the linear model.

5. Predictive Forecasting

5.1. Ten-Year Projections

Based on the evidence from our Bayesian model comparison, which strongly favors the quadratic trend model, we have projected annual atmospheric CO₂ concentrations for the next decade (2026–2036). The quadratic fit confirms that CO₂ accumulation is not merely constant but compounding. As illustrated in Figure 12, the projected trajectory maintains the observed upward curvature, indicating that the annual rate of CO₂ increase is expected to grow over the next ten years.

5.2. Uncertainty Quantification

To quantify forecast uncertainty, we derived 95% credible bands from the posterior predictive distribution. These bands were calculated by drawing samples from the joint posterior distribution of the model parameters $(\beta_0, \beta_1, \beta_2)$ and the residual variance. For each year in the forecast horizon, we computed the regression $\mu(t) = \beta_0 + \beta_1 t + \beta_2 t^2$ across all posterior draws, determining the 2.5th and 97.5th percentiles to define the credible interval.

These bands represent the range within which the true underlying mean CO₂ trend is expected to lie with 95% probability. As seen in Figure 12, the width of these bands increases slightly over time, reflecting the propagation of parameter uncertainty into future projections.

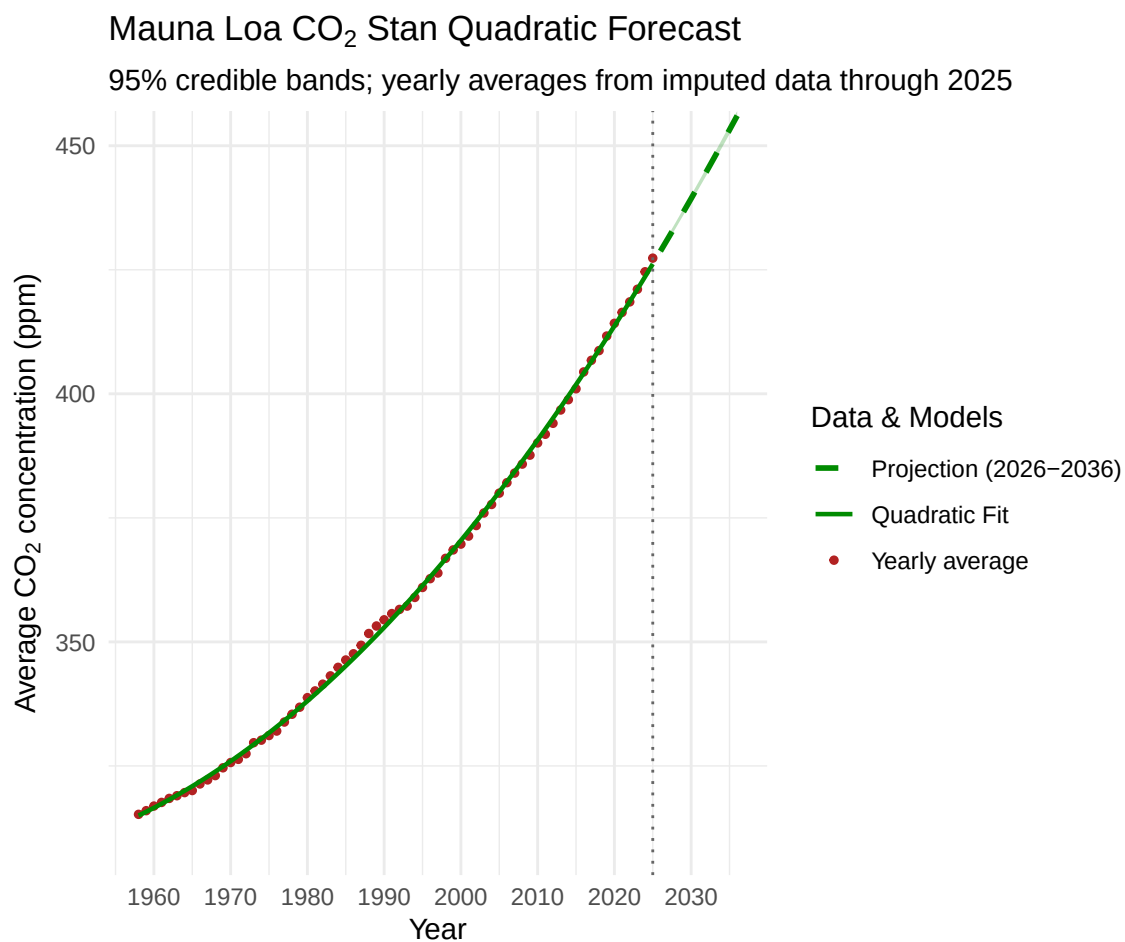


Figure 12: Mauna Loa CO Bayesian Quadratic Forecast. The plot displays the historical yearly averages (red dots), the quadratic model fit (solid green line), and the 10-year projection (dashed green line) with 95% credible bands (shaded region).

Table 9: Projected annual CO₂ concentrations (2026-2036)

Year	Mean Prediction (ppm)	Lower 95% CI	Upper 95% CI
2026	428.76	428.23	429.29
2027	431.36	430.81	431.93
2028	434.00	433.41	434.60
2029	436.66	436.04	437.30
2030	439.35	438.69	440.03
2031	442.07	441.37	442.79
2032	444.81	444.08	445.57
2033	447.59	446.81	448.38
2034	450.38	449.57	451.22
2035	453.21	452.35	454.09
2036	456.06	455.16	456.99

6. Conclusion

6.1. Summary of Findings

Our analysis confirms that the quadratic model significantly outperforms the linear model in capturing the historical CO₂ accumulation at Mauna Loa. The Bayes factor analysis (see Table 7) yielded strong evidence favoring the quadratic model, and the LOO cross-validation (see Table 8) showed a decisive ratio of $|\Delta\widehat{ELPD}|/SE \approx 16.3$, confirming the superiority of the quadratic model. The 10-year projections, based on the quadratic trend, indicate that atmospheric CO₂ will continue to rise at an accelerating rate. These results highlight the non-linear nature of anthropogenic CO₂ accumulation and reinforce the urgency of addressing compounding emissions.

6.2. Limitations and Future Work

While our model robustly captures the historical trend, several limitations should be noted. First, the quadratic model assumes that the current acceleration will persist indefinitely, which may not account for future policy-driven changes in emission rates. Second, the convergence diagnostics suggest that future work could improve model stability by using more informative priors or non-centered parameterizations. Additionally, incorporating exogenous variables such as global economic indicators or policy interventions could enhance predictive accuracy and provide a more nuanced understanding of the drivers of atmospheric CO₂ accumulation.

References